

Ali Alemami

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CS student focused on backend development — systems programming (pipes, processes, threading, shell/network internals) in C/C++, plus hands-on C#.NET application and database development. Seeking an entry-level backend or full-stack role.

EDUCATION

The Hashemite University — BSc in Computer Science 2022 – present

42 Amman — Common Core, Level 4.7/21, 21/24 core projects complete (peer-reviewed, project-based) 2025 – present

TECHNICAL SKILLS

Languages: C, C++, C#, SQL

Systems: Unix & process management, multithreading & synchronization, shell/network internals

CS Fundamentals: OOP, data structures & algorithms, memory management, networking, backend development

Tools: Git, Linux, GDB, Valgrind, Docker, VirtualBox, SQL Server, ADO.NET, .NET Framework

42 CORE CURRICULUM — 21/24 PROJECTS COMPLETE

github.com/ali-alemmami · minishell: github.com/ali-alemmami/minishell · minirt: github.com/ali-alemmami/mini_rt

- **Systems & concurrency (C):** built a POSIX-style shell (minishell) with AST-based command parsing, pipe/redirection execution, and signal-safe process control; implemented the dining philosophers problem (philosophers) with pthreads/mutexes for deadlock-free concurrency; recreated Unix pipeline internals (pipex) with fork/execve/pipe/dup2, eliminating fd leaks via Valgrind
- **Graphics & algorithms (C):** built a raytracer (minirt) with ray-object intersection, lighting, and shadows using MiniLibX; sorted stacks under a limited instruction set (push_swap) using radix sort with custom value indexing
- **Foundations:** CPP Modules 00-09 (OOP, templates, STL), Libft, ft_printf, get_next_line, minitalk, fractol, Born2beroot · in progress: webserv, inception, transcendence

PROGRAMMINGADVICES ROADMAP — 17/24 COURSES COMPLETE

Dr. Mohammed Abu-Hadhoud — algorithms & problem solving, C++, OOP, data structures, C#, SQL Server & T-SQL; currently finishing C# & Database connectivity (ADO.NET)

- **Bank & User Management System (C++, 1,593 lines):** designed an inheritance-based OOP architecture with a bitmask-based role permissions system, custom file serialization for data persistence, and transaction audit logging
- **Contacts Application (C#, ADO.NET, SQL Server):** built a 3-tier console app (Presentation / Business / Data Access layers) with parameterized queries to prevent SQL injection
- **Windows Forms Suite (C#, .NET):** built event-driven desktop apps (Tic-Tac-Toe, pizza ordering) with real-time UI state evaluation and dynamic price recalculation on user input
- **Custom Data Structures (C++):** implemented a doubly linked list with manual memory management for O(1) insertion/deletion at known nodes
- **String & Date Utility Libraries (C++):** built reusable libraries with overloaded static and instance methods, using MSVC property extensions to expose C#-style getter/setter syntax

Also: 5 relational database schema designs (e-commerce, clinic management) · 50+ algorithm problems (Levels 1–6)